4	**(a)**	**(i)**	A decrease in the reading on the top pan balance implies that the net force acting on the balance is now smaller than before. Thus, the force on the magnet due to the current carrying wire is upwards. [B1] By Newton's third law, the force on the wire by the magnet is equal and opposite to the force on the magnet by the wire. [B1] Hence, the force on the wire is downwards. [B1]	**1** **1** **1**

		(ii)	Using Fleming's left hand rule (thumb points downwards, first finger points from North to South) [M1], (second finger points from B to A) current flows from B to A. [A1]	1 1
	(b)	Sinuisoial wave with 2 cycles. [B1] Peaks at + 6.4 mN and –6.4 mN [B1] Time period 25 ms [B1]		1 1 1